1. Clinical Question & Core Hypothesis

Clinical Question

Why do some MIBC tumors respond to intravesical BCG while others do not, and can pretreatment H&E-detectable TLS morphology explain which tumors are ‘primed’ for effective BCG-driven adaptive immunity? 2) Tier 1 path used: the single provided path (Node 1 → Node 2): TLS morphology on H&E → BCG response in MIBC. I build directly on this because it links an H&E-accessible, spatial immune structure to therapy outcome and is immediately testable in retrospective cohorts. 3) Supporting evidence (Tier 2-like rationale): published immunology and pathology findings show that mature TLS (with germinal-center–like architecture) support local antigen presentation, B-cell maturation and T-cell priming in other tumor types, mechanisms plausibly required for BCG-induced tumor control. 4) Endpoint choice: therapy_response (BCG response) was chosen because the clinical question is specifically which patients will benefit from BCG; using a therapy-response endpoint aligns the mechanism (local adaptive priming) with the clinical action (BCG efficacy). 5) Deliberate exclusions: I excluded molecular/genomic features and survival endpoints (OS, PFI) to keep the hypothesis tightly focused on an H&E-only, mechanistic explanation for differential BCG response and to allow direct validation from routine pathology slides; broader multi-omic modifiers and non-therapy endpoints can be explored later as secondary analyses?

Working Hypothesis

BCG efficacy in bladder cancer requires local antigen presentation and B-cell/T-cell priming, and pretreatment histology that shows mature TLS with germinal-center–like architecture and proximal localization to tumor nests on H&E provides the structural substrate for effective BCG-driven immune activation; tumors with high density of mature TLS near tumor areas will thus have higher therapy_response to BCG. Therefore, in MIBC patients treated with intravesical BCG, pretreatment slides demonstrating abundant mature TLS (GC-like) and favorable peritumoral proximity will predict higher therapy_response rates compared with tumors lacking such TLS features.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TLS density (TLS per mm2)	Supporting modifier from knowledge graph
Secondary	TLS maturation proxy (proportion with germinal-center–like pale centers)	Supporting modifier from knowledge graph

Secondary	TLS spatial relation to tumor (intratumoral vs peritumoral vs interface; mean distance to tumor nests)	Supporting modifier from knowledge graph
Secondary	TLS size and area distribution	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TLS density (TLS per mm2): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TLS maturation proxy (proportion with germinal-center–like pale centers): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.